Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

In review ratings column

Customer Data

- Demographics (Age, Gender, Location)
- Subscription Status
- Review Ratings

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Shipping Type

Behavior Metrics

- Discount Applied
- Previous Purchases
- Frequency Patterns

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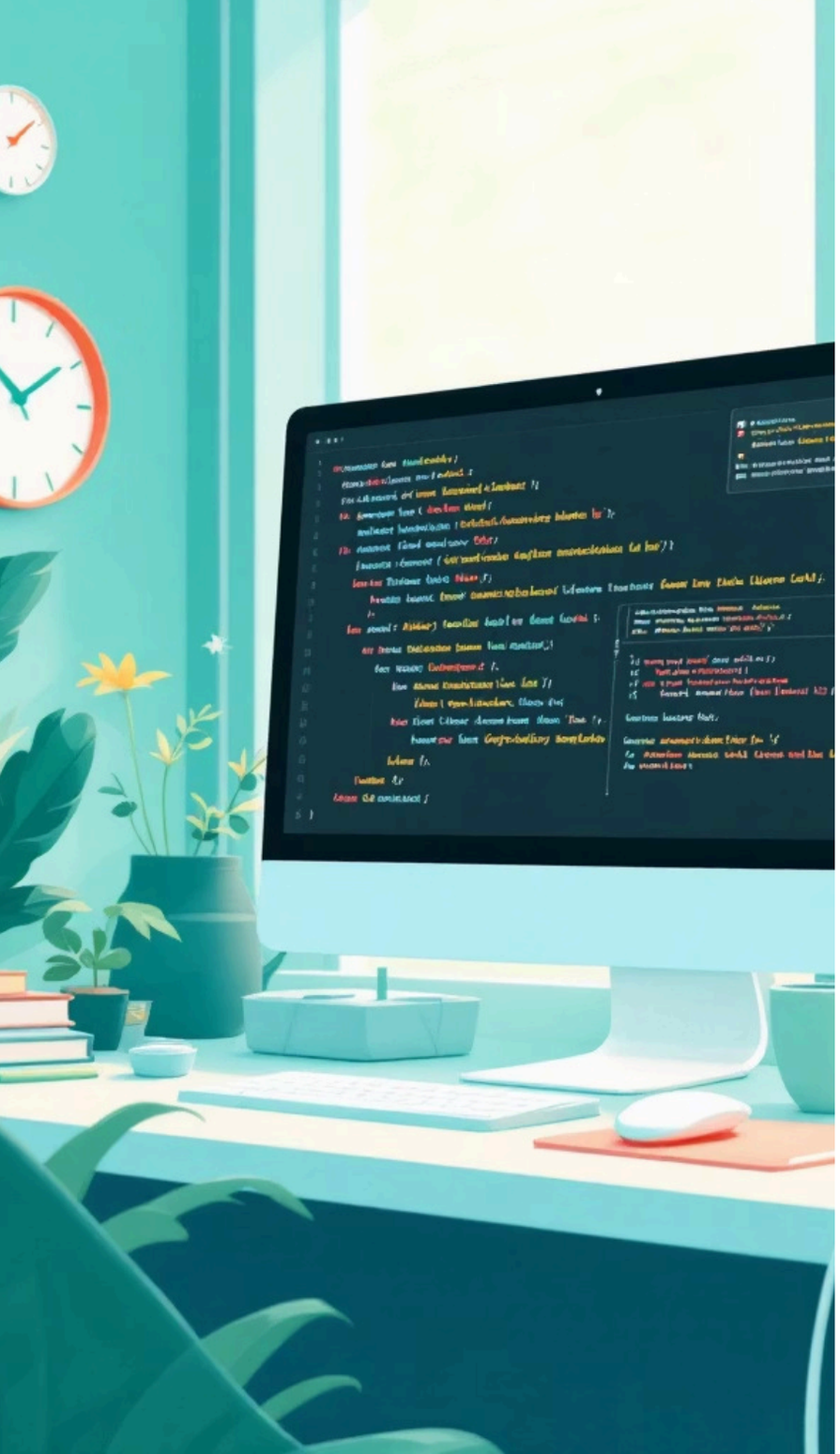

Python Data Preparation

01

Data Loading

Imported using pandas, checked structure with info() and describe()

02

Missing Data Handling

Imputed 37 missing review ratings using median by product category

03

Feature Engineering

Created age groups and purchase frequency days from raw data

04

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame for SQL analysis

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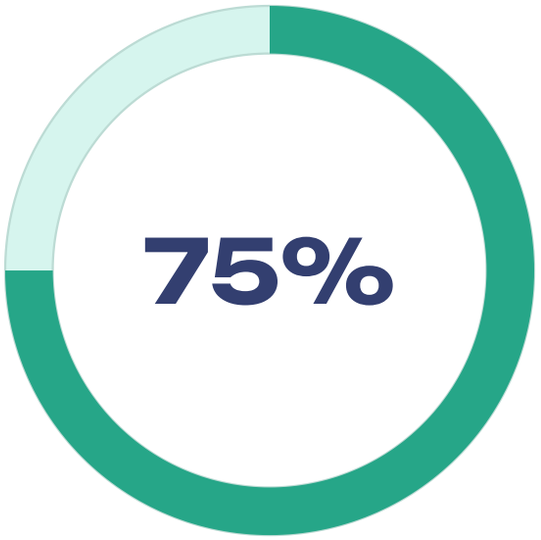
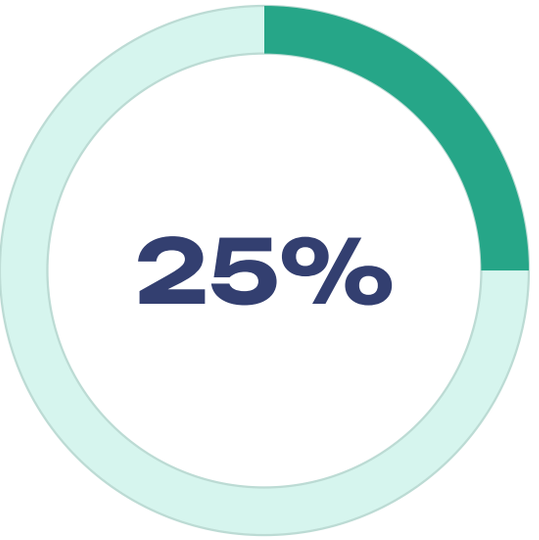
SQL Analysis: Revenue Insights

Revenue by Gender



Male Customers

$157,890 revenue



Female Customers

$75,191 revenue

Shipping Comparison

Standard Shipping

$58.46 avg purchase

Express Shipping

$60.48 avg purchase

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Product Performance

Top Rated

Gloves (3.86), Sandals (3.84), Boots (3.82)

Most Discounted

Hat (50%), Sneakers (49.66%), Coat (49.07%)

Best Sellers

Blouse (171), Pants (171), Jewelry (171)

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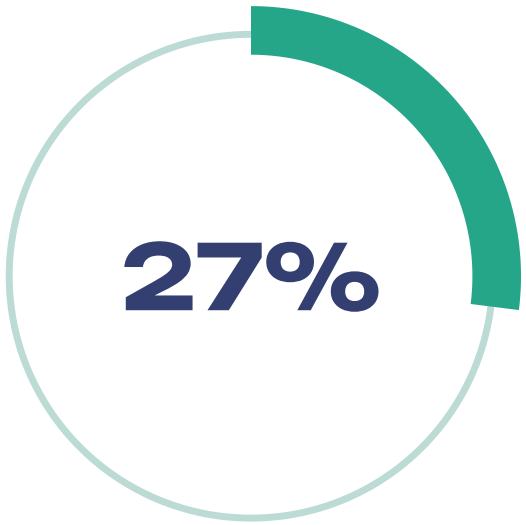
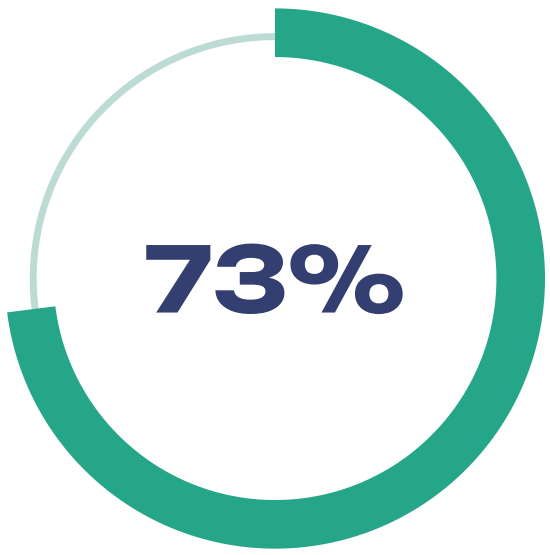
Subscription Analysis



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Average Spend

Subscribers: $59.49
Non-Subscribers: $59.87

Total Revenue

Subscribers: $62,645
Non-Subscribers: $170,436

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Customer Segmentation

Loyal

3,116 customers
High repeat purchase behavior

Returning

701 customers
Multiple purchases recorded

New

83 customers
First-time buyers

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Revenue by Age Group

Young Adult

$62,143 revenue
1,000 purchases

Middle-aged

$59,197 revenue
950 purchases

Adult

$55,978 revenue
900 purchases

Senior

$55,763 revenue
850 purchases

Key Findings

- Male customers drive 3x revenue vs. female customers
- Express shipping customers spend 3.5% more on average
- 958 repeat buyers (over 5 purchases) are subscribed
- Hats, sneakers, and coats see highest discount usage (48-50%)
- Young adults generate highest revenue despite similar purchase frequency

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber base

Customer Loyalty Programs

Reward repeat buyers to move into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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